



A 1920 illustration of the American Lewis and Clark Expedition.

FRENCH CHEMISTS NICOLAS-LOUIS VAUQUELIN (1763–1829) and **Pierre-Jean Robiquet** (1780–1840) isolated **asparagine** from asparagus in 1805. This was the first amino acid (the building blocks of protein) to be identified.

The next year, British inventor **Ralph Wedgwood** (1766–1837) was granted a patent for a type of **carbon paper**. He had originally intended it to help visually impaired people to write, but he later realized it could be used to make duplicates of letters.

In September, the **Lewis and Clark Expedition**, commanded by Captain **Meriwether Lewis** (1774–1809) and Lieutenant **William Clark** (1770–1838) of the US military, reached the Pacific Coast of North America. It had been commissioned by US President Thomas Jefferson to explore the Missouri River after the Louisiana Purchase—the US purchase of 828,000 sq miles (2,100,000 sq km) of French territory—in 1803. The expedition discovered several new species of plants and animals.

In November, British chemist **Humphry Davy** presented his work on the **electrolysis of water**—breaking down water into hydrogen and oxygen by passing an electric current through it (see 1834)—at the Royal Society in London, UK.

“...IT WOULD BE **USELESS** TO ATTEMPT TO **SPECULATE** ON THE **REMOTE CAUSE** OF THE **ELECTRICAL ENERGY**...”

Humphry Davy, British chemist, from *On Some Chemical Agencies of Electricity*, 1806



This engraving shows British chemist Humphry Davy conducting experiments with metals, such as magnesium and barium.

IN 1807, THE NORTH RIVER STEAMBOAT, later renamed *Clermont*, carried passengers from New York City to Albany on the Hudson River in New York, making it the **first commercially successful steamboat**. Designed by American engineer Robert Fulton (1765–1815), the boat completed the 150 mile (240 km) journey in just over 30 hours.

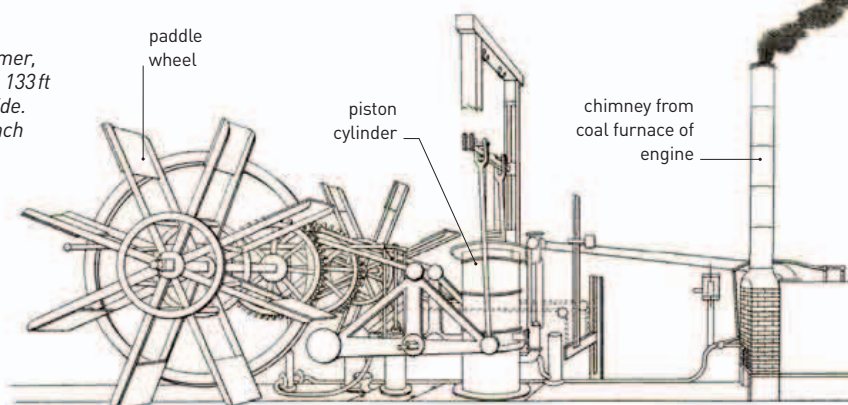
In Britain, chemist **Humphry Davy** used electrolysis to isolate pure forms of many metals, including magnesium, sodium, barium, and calcium. The first metal separated in this way was **potassium**, in 1807.

William Hyde Wollaston patented the **camera lucida** (light room), a drawing aid for artists, in 1807. This device uses a four-sided prism to project an image of the scene to be drawn

First steamboat in the US
Robert Fulton's paddle steamer, later named Clermont, was 133 ft (41 m) long and 12 ft (4 m) wide. It had two paddle wheels, each 15 ft (5 m) in diameter.

onto a drawing surface, allowing the artist to trace it.

Another patent was granted in the same year to French inventors and brothers **Nicéphore** (1765–1833) and **Claude Niépce** (1763–1828) for their invention of an internal combustion engine called the **pyréolophore** (from the Greek words for fire, wind, and bearer). A boat powered by the engine, which burned fine powders such as crushed coal dust, was tested on the Seine River in France.

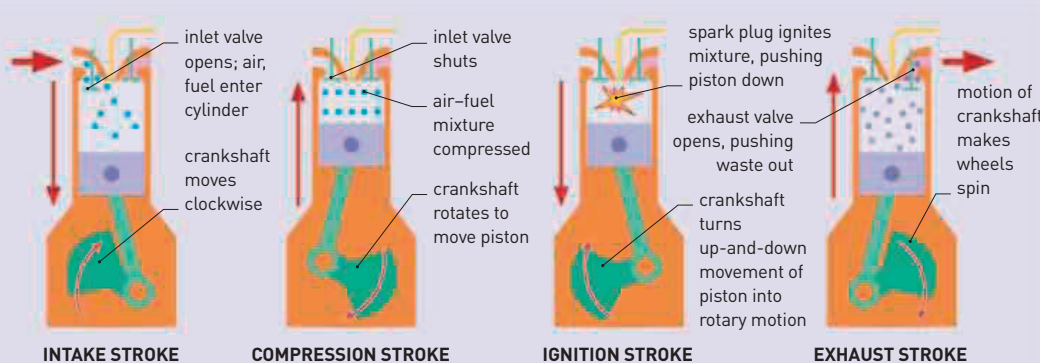


Swiss engineer **François Isaac de Rivaz** (1752–1828) was also working on an engine design and received the patent for a **hydrogen-powered internal combustion engine** in 1807. This early engine used only two strokes. The four-stroke engine (see panel, below) was not developed until 1876.

Unaware of earlier work in the field in Europe, Irish-born American mathematician **Robert Adrain** (1775–1843) published his version of the **method of least squares** in 1808. This statistical technique minimizes the sum of the squares of errors made in a data set, and is used to fit curves (graphs) to data.

INTERNAL COMBUSTION ENGINE

Power is generated in an internal combustion engine by burning fuel within a cylinder. Four-stroke engines operate on a four-stroke cycle. First, a valve lets fuel and air into the cylinder. A piston then moves upward in the cylinder, compressing the air-fuel mixture. A spark plug ignites the mixture, making it explode and pushing the piston down. The piston then pushes the exhaust gases out.



1805 Asparagine, the first amino acid, is isolated

October 7, 1806 Ralph Wedgwood patents carbon paper

September 1806 The Lewis and Clark Expedition reaches the Pacific Coast of North America

November 20, 1806 Humphry Davy reports on the electrolysis of water

1807 William Hyde Wollaston patents the camera lucida

1807 Humphry Davy begins to isolate several metals using electrolysis

1807 François Isaac de Rivaz patents a hydrogen-powered internal combustion engine

July 20, 1807 French inventors the Niépce brothers are granted a patent for an internal combustion engine that runs on powdered coal

August 17, 1807 The first commercially successful steamboat, later named *Clermont*, begins its inaugural journey

February 11, 1808 American Judge Jesse Fell succeeds in burning anthracite, which becomes the fuel of the American Industrial Revolution

1808 French chemist Joseph Louis Gay-Lussac discovers boron

1808 Robert Adrain publishes his version of the method of least squares